KIAN KOK MIDDLE SCHOOL
Second Semester Mock Examination (2026)

Senior Three Commerce

UEC MATHEMATICS
(Paper 2)

MARK

Date: 03 September 2026 (Thu)

(120 mins)

NAME:	NO.	CLASS:

INSTRUCTIONS TO CANDIDATES

1. This subject comprises two papers:
   Paper 1: Multiple-choice Questions (40%),
   Paper 2: Subjective Questions (60%).
2. Paper 2 consists of two sections:
   Section A (Compulsory) : Attempt **all** the five questions (20%)
   Section B (Elective) : Attempt any **four,** but not more than four from the seven questions (40%).
   You are required to attempt a total of **nine** questions.
3. **Begin each question on a fresh page.**
4. Use only **blue** or **black** ink to write your answers and use pencil for drawing only.
5. Do not copy the questions, but the question numbers should be clearly written.
6. Show all mathematical working clearly. Geometric figures should be drawn where necessary.
7. Unless otherwise specified, the prescribed electronic calculators may be used.
8. Arrange the answer scripts in numerical order.
9. Mathematics Formula Sheet is on page 2-3.
10. This paper consists of 7 printed pages.

Do Not Turn Over This Page Until You Are Told To Do So

Prepared by:	Checked by:	Approved by:
(Mr. Adam Pau)	(Mr. Goh Tien Chan)	(Mr. Ong Kin Lun)
Date : 01-09-2026	**Date : 01-09-2026**	**Date : 01-09-2026**

Mathematics Formula Sheet

I. Algebra

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$(a\pm b)^2=a^2\pm 2ab+b^2$$

$$(a\pm b)^3=a^3\pm 3a^2b+3ab^2\pm b^3$$

$$a^2-b^2=(a+b)(a-b)$$

$$a^3\pm b^3=(a\pm b)\left(a^2\mp ab+b^2\right)$$

$$a^m\times a^n=a^{m+n}$$

$$a^m\div a^n=a^{m-n}$$

$$\left(a^m\right)^n=a^{mn}$$

$$(ab)^n=a^nb^n$$

$$\left(\frac{a}{b}\right)^n=\frac{a^n}{b^n}$$

$$\log_a xy=\log_a x+\log_a y$$

$$\log_a\frac{x}{y}=\log_a x-\log_a y$$

$$\log_a x^m=m\log_a x$$

$$a^{\log_a x}=x$$

$$\log_a x=\frac{\log_b x}{\log_b a}$$

Inverse of a matrix $\frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$

A.P. $a_n=a+(n-1)d$

$$S_n=\frac{n}{2}\left[2a+(n-1)d\right]$$

G. P. $a_n=ar^{n-1}$

$$S_n=\frac{a\left(1-r^n\right)}{1-r}$$

II. Trigonometry

Arc length $=r\theta$

Area of a sector $=\frac{1}{2}r^2\theta$

$$\tan\theta=\frac{\sin\theta}{\cos\theta}$$

$$\operatorname{cosec}\theta=\frac{1}{\sin\theta}$$

$$\sec\theta=\frac{1}{\cos\theta}$$

$$\cot\theta=\frac{1}{\tan\theta}$$

$$\sin^2\theta+\cos^2\theta=1$$

$$1+\tan^2\theta=\sec^2\theta$$

$$1+\cot^2\theta=\operatorname{cosec}^2\theta$$

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$$

$$a^2=b^2+c^2-2bc\cos A$$

Area of a triangle $=\frac{1}{2}ab\sin C$

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III. Analytic Geometry

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

Division of a line segment $\left(\frac{mx_2+nx_1}{m+n},\frac{my_2+ny_1}{m+n}\right)$

Area of a triangle $=\frac{1}{2}\left|(x_1y_2+x_2y_3+x_3y_1)-(x_2y_1+x_3y_2+x_1y_3)\right|$

Equation of a line $y-y_1=m(x-x_1)$

IV. Statistics and Probability

Mean $\bar{x}=\frac{\sum f_i x_i}{\sum f_i}$

Mode $=L+\left(\frac{d_1}{d_1+d_2}\right)C$

Median $M=L+\left(\frac{\frac{n}{2}-F_m}{f_m}\right)C_m$

Upper quartile $Q_3=L_3+\left(\frac{\frac{3n}{4}-F_3}{f_3}\right)C_3$

Lower quartile $Q_1=L_1+\left(\frac{\frac{n}{4}-F_1}{f_1}\right)C_1$

Interquartile range $=Q_3-Q_1$

Quartile deviation $Q.D.=\frac{Q_3-Q_1}{2}$

Variance $\sigma^2=\frac{\sum(x_i-\bar{x})^2 f_i}{\sum f_i}=\frac{\sum x_i^2 f_i}{\sum f_i}-\bar{x}^2$

Standard deviation $\sigma=\sqrt{\frac{\sum(x_i-\bar{x})^2 f_i}{\sum f_i}}$

$$=\sqrt{\frac{\sum x_i^2 f_i}{\sum f_i}-\bar{x}^2}$$

Mean deviation $=\frac{\sum|x_i-\bar{x}|f_i}{\sum f_i}$

Statistical index $I=\frac{Q_1}{Q_0}\times 100$

Composite index $=\frac{\sum w_i x_i}{\sum w_i}$

V. Calculus

$$\frac{d}{dx}(uv)=u\frac{dv}{dx}+v\frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$$

$$\frac{d}{dx}x^n=nx^{n-1}$$

$$\frac{dy}{dx}=\frac{dy}{du}\frac{du}{dx}$$

$$\frac{d}{dx}f(g(x))=f'(g(x))g'(x)$$

$$\int x^n\,dx=\frac{x^{n+1}}{n+1}+C,\quad n\neq -1$$

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Section A: Compulsory Questions (20%)

(Attempt all the five questions in this section)

1.

Given that $f(x)=px^2-5x+\frac{1}{x}$, $f'(x)=6x-q-\frac{1}{x^2}$ and $f''(x)=6+\frac{r}{x^3}$, find the values of $p+q+r$. (4%)

2.

(a) Given that $f(x)=\frac{5x+8}{x+3}, x\neq -3$ and $g(x)=\sqrt{7x-3}$. Find the value of $g\circ f(-4)$. (2%)

(b) If $f(x)=px-5x^2-13x^3+6x^4+7$ is divisible by $3x+1$, find the value of p. (2%)

3.

(a) Express the quadratic function $f(x)=x^2+5x+8$ in the form $f(x)=(x+p)^2+q$. Find the value of $p-q$. (2%)

(b) Two coins are rolled at the same time, find the probability of getting at most one head? (2%)

4.

Given that $y=\frac{5}{(3x-1)^2}$ and $\frac{dy}{dx}=-8g(x)$. Find $\int_0^2[3x+g(x)]dx$. (4%)

5.

From a plane flying due east at $285m$ above sea level, the angles of depression of two ships sailing due east measure 36° and 26°. How far apart are the ships? (Give your answer round to the nearest tenth) (4%)

Section B: **Elective Questions (40%)**

(*Attempt any **four**, but **not more than** four from the seven questions in this section*)

6.

(a) Given that $A=\begin{pmatrix}-3 & 5\\ 4 & -3\end{pmatrix}$, find $\left(A^T\right)^{-1}$. Hence, solve the system of equations

$$\begin{cases}-3x+4y=22\\ 5x-3y=55\end{cases}.$$ (4%)

(b) The table shows the items needed for a bowl of meat noodles. The index number in the year 2025 based on the year 2023 and their expenditure are shown in the table:

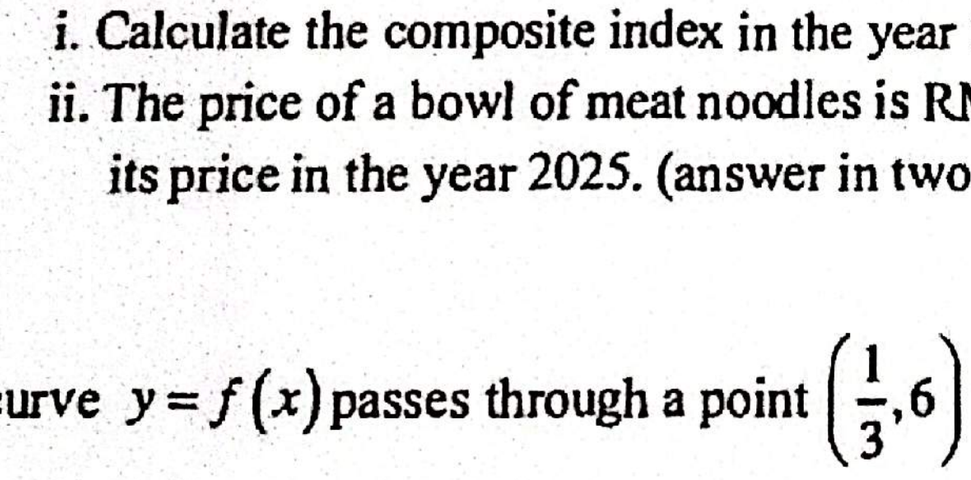
Item	Index number	Expenditure
Gas	102	20%
Cooking oil	120	13%
Meat	115	27%
Vegetable	107	7%
Noodles	110	33%

i. Calculate the composite index in the year 2025 based on the year 2023.

ii. The price of a bowl of meat noodles is RM4.00 in the year 2023, find its price in the year 2025. (answer in two decimal places) (2%)

(c) A curve $y=f(x)$ passes through a point $\left(\frac{1}{3},6\right)$ .If the gradient of tangent at a point is $\frac{dy}{dx}=5-\frac{2}{x^2}$. Find the equation of the curve. (4%)

7.



(a) Find the coordinates of the points on the curve $y=-5x^3-5x$ such that the gradient of the normal at this point is $\frac{1}{65}$. (4%)

(b) Diagram below shows a quadrilateral $ABCD$. The area of the triangle ABD is $16cm^2$ and $\angle BAD$ is an acute angle. Find $\angle CBD$. (6%)

8.

(a) Given the sum of the first n terms of an arithmetic progression is $S_n=\frac{3n^2+7n}{2}$. Find the value of h if $S_h=1025$. Hence, calculate the hth term. (5%)

(b) Given a quadratic equation $mx^2+6x+3m=0$ .Find the value of m if

i. the sum of roots of the quadratic equation is 12;

ii. the difference between roots of the quadratic equation is 2. (5%)

9.

(a) Given that $f(x)=3x-2$, $g(x)=5x^2-2x+3$ and $(f\circ g)(a)=g(3)-f(-5)-4$ .Find the value of a. (5%)

(b) (i) Solve the equation $\frac{3}{x+2}-\frac{2}{x-2}=\frac{2x-5}{x^2-4}$. (2%)

(ii) Solve the inequality $(x+8)(x-9)\geq(x+8)$. (3%)

10.

(a) If $P(x,5)$ divides externally $A(-7,0)$ and $B(8,y)$ in the ratio 3:8, find the values of x and y. (4%)

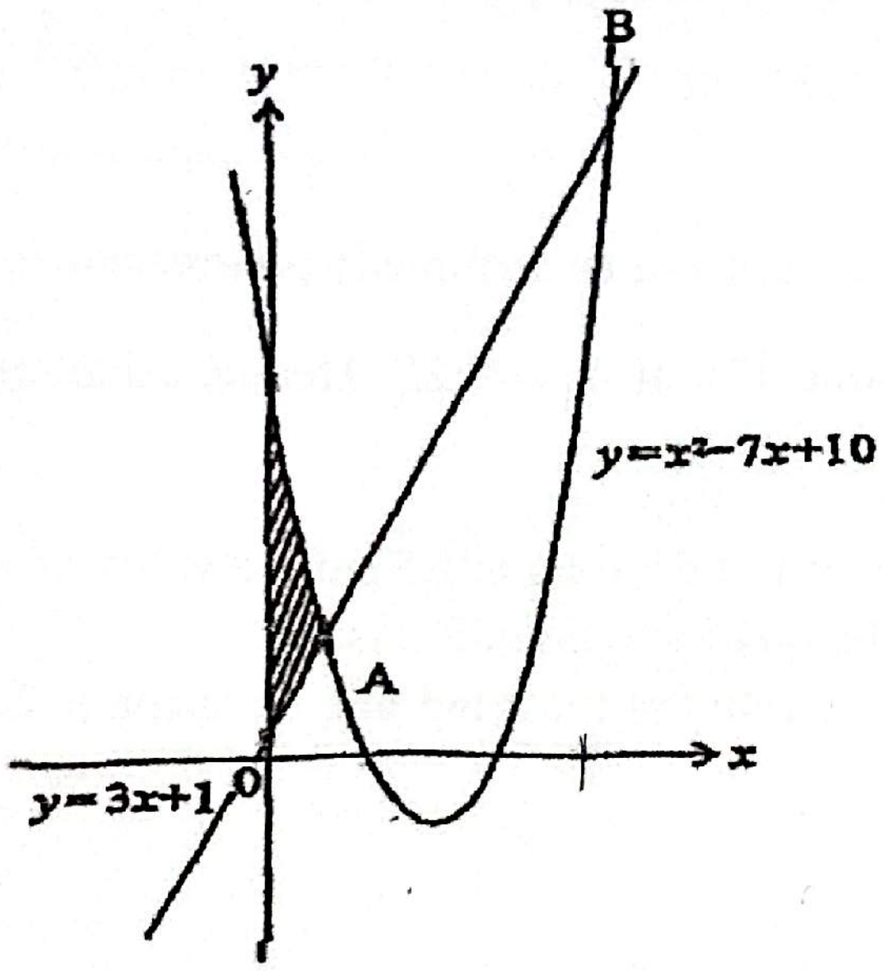
(b) Given that $\tan A=\frac{1}{2}$, where A is in Quadrant *III*. Find the value of $\cot A+\sin A+\sec A$. (3%)

(c) Solve $3\cdot 5^{x+2}=8^{x-3}$. (Give the answer correct to two decimal places) (3%)

11.

(a) Find the standard deviation of the data 2,3,5,7,8. Then multiply each data by 4. Find the standard deviation of the new values. (3%)

(b) Find the area of the shaded region in Fig below.



(4%)

(c) Given that three points $(-5,8),(a,-2),(-3,12)$ are collinear, find the value of a. (3%)

12.

The normals to the curve $y = x^3 - 5x^2 + 3x + 1$, at the points $A(4,-3)$ and $B(1,0)$, meet at the point C.

i. Find the coordinates of C; (4%)

ii. Find the area of triangle ABC. (3%)

(b) The second term of a geometric progression is 9 less than the first term. The sum of the second and third terms is 30. Given that all the terms in the progression are positive, find the first term. (3%)

END OF QUESTION PAPER